

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The network analyzer tool, tcpdump log shows that the udp port 53 is unreachable. Port 53 is used primarily for UDP in DNS protocol. In the error message, "udp port 53 unreachable". The word "unreachable" in the message indicates the UDP message requesting an IP address for the domain "www.yummyrecipesforme.com" did not go through to the DNS server because no service was listening on the receiving DNS port. This may indicate a problem with the DNS server or firewall configurations. It is possible that this is an indication of a malicious DDoS attack on the DNS server.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

The incident was reported in the afternoon when several customers of clients reported that they were not able to access the client company website www.yummyrecipesforme.com, and saw the error "destination port unreachable" after waiting for the page to load. The network security team responded and began running tests with the network protocol analyzer tool tcpdump. The resulting logs revealed that port 53, primarily used by UDP in DNS protocol, is unreachable. We are continuing to investigate the root cause of the issue to determine how we can restore access to the website. Our next steps include checking the Firewall configuration to see if port 53 is blocked and contacting the system administrator of the DNS server to check the system for signs of an DDoS attack. The security team believes this may be the work of malicious actors trying to hamper the business operations of the company.